

DIANA ALVAREZ SANCHEZ

Hidalgo, TX

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Education

University of Texas Rio Grande Valley

Aug 2022 – Dec 2025

Bachelor of Science in Computer Science

Edinburg, TX

Relevant Coursework

- Data Structures
- Algorithms
- Computer Networks
- Digital Systems
- Cloud Computing
- Cybersecurity
- Deep Learning in Medical Imaging
- Database Management
- Web App Development

Projects

Language Learning App | *React, Django, Gemini AI, GitHub, Figma*

Sep 2025 - Dec 2025

- Developed an AI-driven web app using React and Django to analyze media and deliver accurate translations and linguistic feedback.
- Integrated Gemini AI to enhance contextual accuracy, providing culturally relevant definitions over standard translation apps.
- Managed an Agile development cycle with a 4-member team using GitHub and Figma to deliver a cross-media prototype.

Secure VPN Gateway Deployment | *OpenVPN, Microsoft Azure, Ubuntu*

Nov 2025 - Dec 2025

- Deployed a secure OpenVPN gateway on Microsoft Azure with Ubuntu, enabling encrypted remote access via a private subnet.
- Implemented PKI-based authentication and TLS encryption, debugging certificate issues to ensure reliable connectivity.

Brain Tumor 3D Segmentation | *Python, PyTorch, MONAI*

Feb 2025 - May 2025

- Reimplemented UNETR and SegResNet using Python, PyTorch, and MONAI to automate segmentation of 750 3D MRI volumes.
- Achieved high segmentation accuracy with SegResNet (Dice = 0.87), streamlining diagnostic imaging and treatment planning.

Task Manager Web & Desktop App | *Python, Tkinter, Django, SQL, JavaScript*

Feb 2025 - May 2025

- Developed a task manager desktop (Python/Tkinter) and web app (Django/SQL) for individual and admin-controlled tracking.
- Built role-based interfaces with a responsive HTML/CSS/JavaScript front-end, letting admins manage assignments and users securely update task statuses.

Technical Skills

Languages: C++, Python, Java, Rust, HTML/CSS, JavaScript, JSX, SQL, C

Developer Tools: VS Code, Git, Docker, Figma, Google Cloud Platform

Technologies/Frameworks: React, Django, Gemini API, MONAI, Linux/Unix, UI/UX

Hardware/Embedded: Raspberry Pi, Arduino, Microcontrollers

Work Experience

H-E-B Grocery

Sep 2023 – Present

Personal Shopper

Hidalgo, TX

- Managed multiple simultaneous curbside pickup orders under strict deadlines while resolving customer inquiries to ensure a positive shopping experience.

Leadership / Extracurricular

Society of Women Engineers

Oct 2024 – Present

Member

- Participated in workshops and events to advance the inclusion and representation of women in engineering.

ColorStack

Dec 2025 – Present

Member

University of Texas Rio Grande Valley

- Engaged with a national community of Black and Latinx CS students through technical workshops, peer mentorship, and professional networking.

Frontera Devs

Jan 2023 – Dec 2025

Member

University of Texas Rio Grande Valley

- Strengthened technical and collaboration skills through workshops and team development events.